PROBLEM STATEMENT

Search and rescue and emergency response operations are significantly hindered by low-light and no-light conditions. Limited visibility reduces situational awareness, slows mobility, and increases the risk to both personnel and victims. In dynamic field environments—such as disaster zones, infrastructure inspections, and outdoor recovery missions—uncertainty in locating individuals directly impacts safety and operational effectiveness. Current illumination and visual assistance technologies do not adequately address these constraints, revealing a gap between operational demands and available solutions.

Need Statement

Many tasks in search and rescue and emergency response operations occur in remote, rugged terrain with limited visibility where individuals may become lost or injured. Search and rescue operations in wilderness areas often become more difficult as daylight fades, increasing risk to both missing persons and first responders [1]. Search efforts are particularly time-sensitive; criminology experts note that the likelihood of locating a missing person decreases rapidly after the first 72 hours of an investigation [2]. Traditional search strategies rely on last-known positions and manual search patterns coordinated through emergency operations centers; these methods can be slow, resource-intensive, and constrained by visibility and terrain challenges [3]. Existing mobile and fixed lighting solutions constrain operator mobility and provide limited, non-adaptive illumination coverage. These limitations contribute to reduced situational awareness, increased physical fatigue, and higher likelihood of operational errors. Fixed lighting systems lack portability and adaptability, while vehicle-mounted lighting cannot effectively illuminate confined or obstructed areas. A solution is needed to improve visibility and situational awareness in low-light environments while preserving operator mobility and minimizing operational strain.

Objective Statement

The objective of this project is to design and prototype a portable aerial lighting and visual assistance system that enhances visibility and situational awareness in low-light and no-light operational environments. The Aerial Matrix will provide hands-free illumination and visual support for personnel engaged in search and field operations. It will enable coordinated operation among multiple aerial units to extend illumination coverage across complex terrain. The system supports modular lighting and visual sensing payloads while maintaining portability and ease of deployment for small teams.

Background and Related Work

Studies on missing person investigations emphasize the time-critical nature of early search phases, with the first 72 hours being crucial before leads diminish and search complexity increases [2]. Emergency operations centers leverage technologies such as GPS and mobile communications

when available; however, these tools are not universally effective in remote or obstructed environments [3]. Existing portable lighting solutions—including handheld flashlights, helmet lights, and temporary floodlights—provide localized illumination but restrict mobility and limit environmental awareness. Aerial systems capable of delivering dynamic, adaptive lighting and visual sensing offer the potential to reduce search times and expand coverage; however, current implementations remain constrained by operational complexity, control demands, and resource requirements.

2. DESIGN REQUIREMENT SPECIFICATIONS

This portion of the document will provide specifications that enable *Aerial Matrix*™ to function as search and rescue solution that is portable, adaptable, and offers provides good visibility. These specifications outline engineering limitations, operational requirements, marketing considerations, and other relevant factors that influence the system design.

2.1 Features and Requirements

The *Aerial Matrix*™ will be an adaptable system capable of traversing a wide variety of rough terrain. Using object detection, the user will have access to a flexible targeting system. A user-friendly interface will allow for a smooth selection of targets. With synchronized flight, the *Aerial Matrix*™ will be scalable for all types of Search and Rescue (S&R) operation. These features are supported by system design requirements as shown in Table 2-1.

2-1 Features and System Design Requirements

Features	System Design Requirements	Justification
Flexible Targeting	Target identification and tracking shall be performed using an Object Detection of at least 3 targets.	Object detection allows the drone to follow a given object intelligently.
Modular Attachments	Attachments capable of being interchanged with one hand within 20 seconds.	This will provide a standardized platform for new attachments and ease of deploying them.

Scalable Deployment	Drones must be able to communicate between at least 3 drones.	This will enable flexible numbers of drones and locations.
Synchronized Self-Flight	Staying in the assigned position within a range of ±0.5m based on parent drone commands.	Allow stable and performant intelligent flight control without direct and constant user input.
User Friendly Interface	A mobile application shall provide deployment configuration navigation accessible within 10 seconds of home screen.	This will allow the user to configure the drone swarm visually and provide selections for targets.

Justification

The proposed system emphasizes modularity, scalability, autonomy, and usability to address limitations in existing drone solutions. A standardized modular attachment system allows rapid payload changes, improving flexibility across missions such as search and rescue or inspection. Mesh-based communication enables scalable deployment and increases reliability by allowing drones to relay data without dependence on a single node. Incorporating object detection supports autonomous targeting and tracking, reducing the need for constant user control, and improving response time in complex environments. A mobile application interface simplifies user interaction by providing real-time visualization and intuitive controls, while centralized orchestration of flight ensures stable and coordinated swarm behavior. Together, these features create a system that is adaptable, efficient, and aligned with modern advancements in multi-drone and autonomous systems

2.1.2 Subsystem Requirements and Demonstration Setup

Table 2-2 summarizes the major subsystems, their design requirements, and the corresponding demonstration methods used to validate system performance. Each requirement is paired with a measurable test to ensure functionality under expected operating conditions.

Table 2-2: Subsystem Requirements and Demonstration Setup

Subsystem/ Item:	Requirements	Demonstration Setup
Drone Attachments	- Attachments capable of being interchanged with one hand within 20 seconds. - Attachment mounting must be able to withstand the drone dropping from a height of 10m without dislodgement or interruption of function.	Interchanging functionally different attachments by hand with demonstration of attachment capability after drop from a measured height.
Drone Construction	- Drone construction must stay under 1kg. - Must be able to fly with a 250g payload.	Demonstrate weight via scale. Carry known weight during preset flight path.
Drone Controls	- Provide stable hovering, staying in the same position (±0.25m) for more than 5 minutes. - Providing time series database for scraping data from each drone & attachment every 10s.	Provide drone flight with a visually stable appearance. Give positional changes via laptop CLI.
Drone Network	- Drones must be able to communicate over a distance of at least 15 meters. - Must be able to transmit TCP packets within 500ms with a bandwidth of at least 5megabit.	Ping and run bandwidth benchmark while drones are separated by 15m.
Object Detection	- The camera must be able to track at least 3 objects of interest of size of at least 30in x 30in with 65% accuracy and within 5 seconds. - The Camera should be able to detect objects from 2 to 10 meters from above.	Show model detecting objects and highlighting its "target" from an over-head position.

2.2 Constraints

2.2.1 Constraints Preview

Table 2-3 outlines the key constraints impacting the system, including cost, time, manufacturability, and environmental conditions. These constraints define the practical limits within which the system must be designed and operated.

Table 2-3: Constraints

Type	Name	Description
Economic	Cost	The project requires components that could exceed the allotted budget of $1000.
Economic Complexity	Time	Due to the nature of the project and the new, complicated, and uncharted territory, the project will take time and patience to reach its full potential over the span of two semesters.
Manufacturability	Weight	Because the project is housed in the drones, it will have to guarantee that both parts are small and light, and the motors are strong enough for lift.
Environment	Durability	Since these drones will be used outside with varying conditions. Our system will be designed to operate in winds below 10mph and light rain.
Power	Flight Time	With these drones being used actively for S&R, a minimum flight time of at least 20 minutes is required for general operations.

2.3 Standards

Table 2-4: Standards

Specific Standard	Standard Document	Specification/Application
ASTM F2910	This product meets the requirements of being under a gross weight of 25 kilograms and is allowed to be designed under the ASTM F2910 standard.[4]	International standard for unmanned aircraft under 25kg.
CFR Title 14, Chapter I, Subchapter C, Part 48	This part provides registration and identification requirements for small, unmanned aircraft.[5]	United States of America's FAA requirements for unmanned aircraft under 25kg.
IEEE 802.11	This product uses a Wi-Fi wireless network for communication under the standard IEEE 802.11.[6]	Enabled wireless communication of ~140m outside over 2.4GHz

References

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[2] J. Jacobo, "Why the first 72 hours in a missing persons investigation are the most critical, according to criminology experts," ABC News, Oct. 8, 2018. [Online]. Available: **https://abcnews.go.com/US/72-hours missing-persons-investigation-critical-criminology-experts/story?id=58292638**

[3] "Locating a missing person in a hostile environment: tools and skills of an Emergency Operations Room," Rescue Press Academy. [Online]. Available: **https://academy.rescue.press/locating-a-missing-person-in hostile-environment-tools-and-skills-of-an-emergency-operations-room/**

[4] ASTM International, *F2910 Standard Specification for Design and Construction of a Small Unmanned Aircraft System (sUAS)*. West Conshohocken, PA, USA: ASTM.

[5] Electronic Code of Federal Regulations, "Title 14, Chapter I, Subchapter C, Part 48 – Registration and Marking Requirements for Small Unmanned Aircraft," *eCFR*. [Online]. Available: **https://www.ecfr.gov/current/title-14/chapter-I/subchapter-C/part-48**.

[6] IEEE Standards Association, "The Evolution of Wi-Fi Technology and Standards." [Online]. Available: **https://standards.ieee.org/beyond-standards/the-evolution-of-wi-fi-technology-and-standards/**.